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Compound Interest

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Compound Interest

Compound Interest Definition:

Compound interest is the interest calculated on the principal and the interest accumulated over the previous period. It is unlike simple interest where **interest** is not **added to the principal** while calculating the interest during the next period. Compound interest finds its usage in most of the transactions in the banking and finance sectors and also in other areas as well. Some of its applications are:

- Increase or decrease in population.
- The growth of bacteria.
- Rise or depreciation in the value of an item.

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Compound Interest Formula:

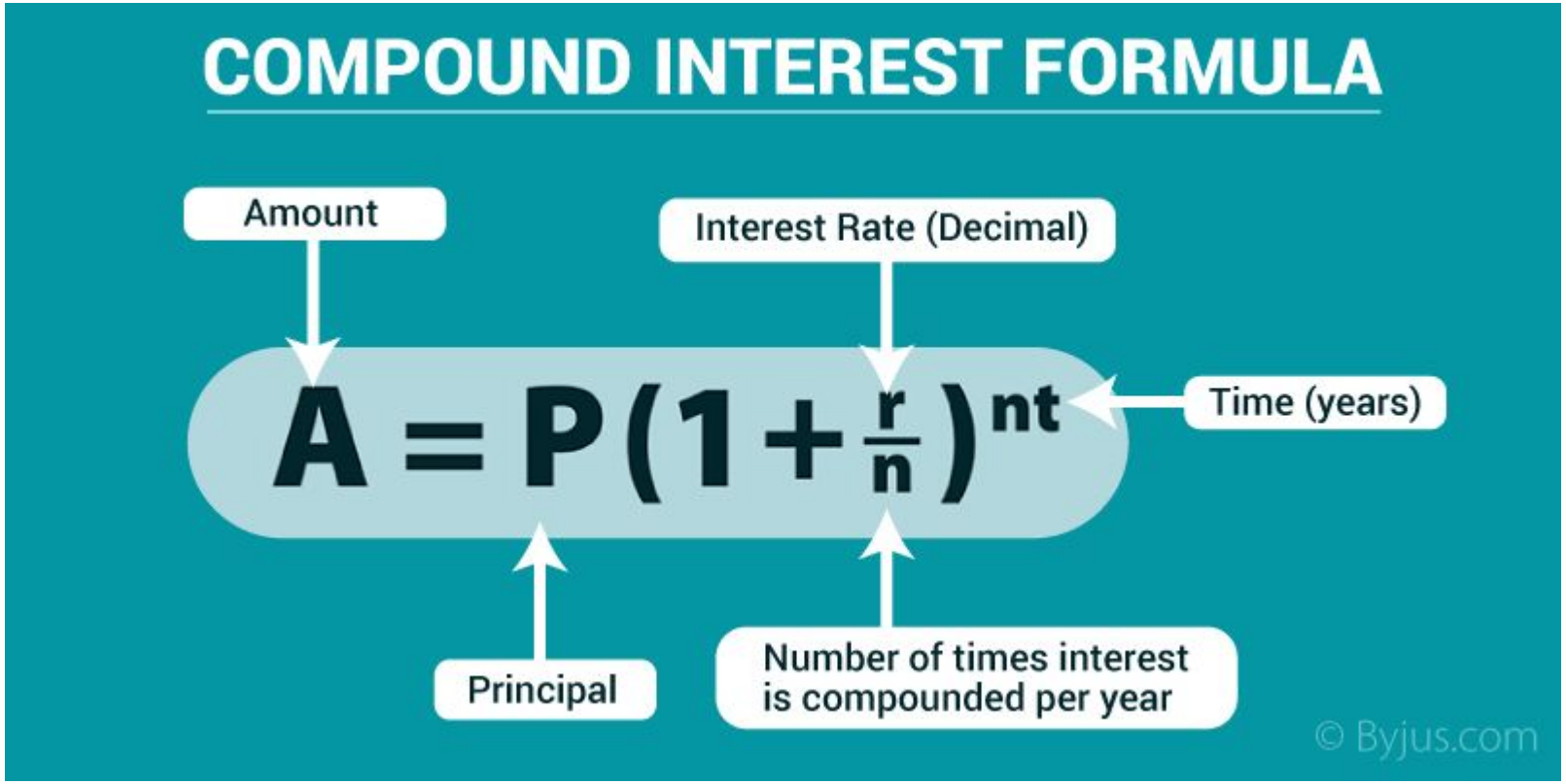
The compound interest formula is given below:

Compound Interest = Amount – Principal

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Where the amount is given by:



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Where,

A= amount
P= principal
R= rate of interest
n= number of years

It is to be noted that the above formula is the general formula for the number of times the principal is compounded in an year. If the amount is compounded annually, the amount is given as-

$$A=P(1+(R/100))^t$$

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If interest is not compounded yearly then,

Compounded Half Yearly	Compounded Quarterly
• Interest is added after every six months to form a new principal	• Interest is added after every three months to form a new principal
• Rate of Interest is equal to half of annual rate i.e. $R = R/2$	• Rate of Interest is equal to one-fourth of annual rate i.e. $R = R/4$
• Since in one year there are two half years, so $T = 2T$	• Since in one year there are four quarters, so $T = 4T$
• Amount $= P\left(1 + R/200\right)^{2T}$	• Amount $= P\left(1 + R/400\right)^{4T}$
• CI $= P\left[\left(1 + R/200\right)^{2T} - 1\right]$	• CI $= P\left[\left(1 + R/400\right)^{4T} - 1\right]$

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Question 1

Find the compound interest on Rs.8000 at 5% per annum for 3 years when C.I is reckoned yearly.

A) Rs. 1262

B) Rs. 1261

C) Rs. 1260

D) Rs. 1361

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Question 2

A sum is invested at compound interest payable annually. The interest in the first two successive years was Rs. 400 and Rs. 420. The sum is _____.

A) Rs. 8000

B) Rs. 7500

C) Rs. 8500

D) Rs. 6500

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Question 3

At what rate percentage per annum will Ron lends a sum of $ 2000 to Ben. Ben returned after 2 years $ 2205, compounded annually?

A) 2%

B) 3%

C) 4%

D) 5%

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Question 4

Find the compound interest on Rs. 500 for 1 year at 40% per annum compounded quarterly.

A) Rs. 146.01

B) Rs. 133.1

C) Rs. 732.50

D) Rs. 232.05

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Question 5

The annual sales of a company in the year 2000 was Rs. 1000 and in the year 2005 was Rs. 2490. Find the compounded annual growth rate of sales in the given period of the same company.

A) 14.28%

B) 125%

C) 17.86%

D) 20%

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Question 6

What is the amount after 3 years on a sum of Rs. 15638 being compounded annually at interest of 10% per annum ?

A) Rs. 20814

B) Rs. 19824

C) Rs. 21924

D) Rs. 19784

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Question 7

Ram purchased a Hyundai car 3 years ago for Rs. 0.4 million. Its value depreciated each year @ 20% p.a. What is the present value of the car?

A) 691200

B) 241800

C) 204800

D) 24800

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Question 10

Reena got a loan of Rs. 12000 against her fixed deposits to purchase a mobile phone. If the rate of interest is 12% p.a. compounded half yearly, then the amount (in Rs.) that she has to pay back after 2 years is _____.

A) 24883

B) 13483

C) 18882

D) 15149

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Question 11

The effective annual rate of compound interest corresponding to a compound interest of 8% per annum payable half-yearly is _____.

A) 8.16%

B) 6.16%

C) 7.16%

D) 5.16%

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Question 12

The amounts for compound interest on a certain sum at a certain rate of interest after the second and third year are Rs. 19860 and Rs. 21846, respectively. What is the rate of interest?

A) 10%

B) 20%

C) 30%

D) 40%

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Question 13

Pravin borrowed a sum of money and returned it in three equal quarterly instalments of Rs. 17576 each. Find the sum borrowed, if the rate of interest charged was 16% per annum compounded quarterly.

A) 46900

B) 48775

C) 68320

D) 44556

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Question 14

Mr. Venu invests Rs. 20000 in an account that pays 4% interest per year, compounded quarterly. What is the amount of money that he will have after 2 years?

A) 21657.12

B) 20282.52

C) 22802.56

D) 22028.58

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Question 15

A certain amount of money is lent out at compound interest at the rate of 40% per annum for two years, compounded annually. It would give Rs. 568 more if the amount is compounded half yearly. The principal is Rs. _____.

A) 10000

B) 12000

C) 6000

D) 5000

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Question 16

The death rate of a town with a population of 5000 is 20%. Considering that there are no new births, what is the population of town after 3 years?

A) 1280

B) 3600

C) 2560

D) 2780

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Question 17

Kishore invests Rs. 3000 for three years at a certain rate of interest, compounded annually. At the end of one year it amounts to Rs. 3540. Calculate the amount due at the end of the second year.

A) 4035.6

B) 4071

C) 4080

D) 4177.2

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Question 18

A bank offers 10% compound interest calculated on half-yearly basis. A customer deposits Rs. 1000 each on 1st January and 1st July of a year. Find the interest he would have gained at the end of a year.

A) Rs. 2152.50

B) Rs. 152.50

C) Rs. 1152.50

D) Rs. 315

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Question 19

The population of goats in a village was 5200 three years back. It is 6600 right now. What will be the population three years down the line, if the rate of growth of population of goats is constant over the years?

A) 6400

B) 8000

C) 8800

D) 6600

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Question 20

A sum of Rs. 10,000 is divided between 2 siblings who are 5 and 6 years old in such a way that if their portions are invested at the rate of 2% per annum compound interest, they shall receive equal amounts on reaching 18 years of age. How much money does the younger one get initially?

A) 4000

B) 4950

C) 4735

D) 4335

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THANK YOU

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